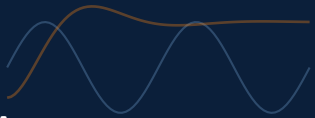
DAY 5 / 5

Mechatronics: Electrical Machines, Control, Servo Systems and Sensing



Resonance, Sensors, Measurement and System Integration

3.7 Resonance & filters ▪ 4.1 Sensors ▪ 4.2 Encoders & resolvers ▪ 4.3 Sampling ▪ 4.4 Signal processing ▪ 4.5 Communication

Mechatronics Training Programme

Theoretical Course Series ▪ Day 5 of 5

Agenda and learning objectives

Code	Session	Time
3.7	Resonance, vibration and filters	60 min
4.1	Sensors and measurement	30 min
4.2	Encoder and resolver	45 min
4.3	Sampling and digitisation	30 min
4.4	Signal processing and filters	90 min
4.5	Communication and integration	45 min
	Course wrap-up (inside 4.5)	

5 h net teaching time; 10-min breaks roughly every hour are extra.

By the end of today you can . . .

- explain why a flexible shaft makes a servo ring
- calculate a resonance frequency
- explain what a notch filter does
- tell accuracy from precision
- explain how an encoder counts angle
- apply the Nyquist rule to sampling
- write a simple digital filter
- sketch how drives talk on a fieldbus

What we learned so far: Days 1–4

Day 1 Motors turn current into torque: $T = k_t\, i$

Day 2 Models, step response, PID, stability

Day 3 Filters, discrete (sampled) control, observers

Day 4 The drive: FOC, current / velocity / position loops, tuning

Day 5 Real mechanics, real sensors, real networks

Three ideas we reuse today

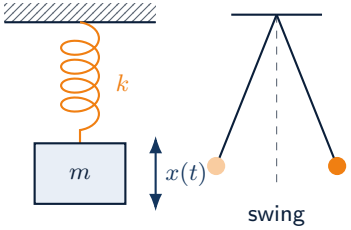
- **Frequency response**: how much a system amplifies a sine of frequency f (gain) and how much it delays it (phase).
- **Bandwidth** f_c: up to which frequency the loop follows commands well.
- **Phase margin**: safety distance from oscillation. Every delay or filter eats some of it.

Days 2–4 assumed a *rigid* load and *perfect* sensors. Today we drop both assumptions.

Session 1

Resonance, vibration and filter use (3.7)

Refresher: natural frequency of a mass on a spring



- Pull and let go: it oscillates at its **natural frequency**.
- A swing has one too: push in rhythm and it builds up.

$$\omega_n = \sqrt{\frac{k}{m}} \ \text{[rad/s]}, \qquad f_n = \frac{\omega_n}{2\pi} \ \text{[Hz]}$$

k spring stiffness [N/m], m mass [kg]. Rotating version: k in N m/rad, J in kg m^2.

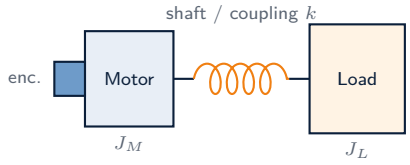
Example

1. $m = 1\,\text{kg}$, $k = 100\,\text{N/m}$
2. $\omega_n = \sqrt{100/1} = 10\,\text{rad/s}$
3. $f_n = 10/(2\pi) \approx 1.6\,\text{Hz}$

- Stiffer spring $\Rightarrow$ higher f_n; heavier mass $\Rightarrow$ lower f_n.
- **Resonance**: driving at f_n gives a large response if damping is small.

Why mechanical resonance limits servo bandwidth

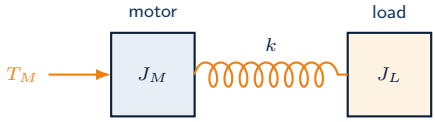
- Days 3–4 assumed a **rigid** load: motor and load move as one.
- Real shafts, couplings, belts and ball screws are **compliant**: they behave like springs.
- Inertia + spring = a mass–spring system with very little damping.
- Raise the controller gain to make the axis stiffer and, at some point, the axis **rings or becomes unstable**.



Typical resonance: belts 20 Hz to 100 Hz, ball screws 100 Hz to 500 Hz, couplings / shafts 0.3 kHz to 3 kHz

Analogy: a car on soft suspension. Steer hard and fast and the body sways; the driver (controller) must not excite that sway.

The two-mass model and its two special frequencies



$$\omega_{\mathrm{AR}} = \sqrt{\frac{k}{J_L}}, \qquad \omega_{\mathrm{R}} = \omega_{\mathrm{AR}}\sqrt{1+R}$$

$$R = J_L/J_M \text{ (inertia ratio)}$$

- **Anti-resonance** (AR): load swings, motor stands still.
- **Resonance** (R): motor and load swing against each other.

Worked example

1. $J_M = 1.2 \times 10^{-4}\,\mathrm{kg\,m^2}$, $J_L = 6 \times 10^{-4}\,\mathrm{kg\,m^2}$, $k = 600\,\mathrm{N\,m/rad}$
2. $\omega_{\mathrm{AR}} = \sqrt{600/(6 \cdot 10^{-4})} = 1000\,\mathrm{rad/s} \Rightarrow f_{\mathrm{AR}} = 159\,\mathrm{Hz}$
3. $R = 6/1.2 = 5$, $\sqrt{1+R} = 2.45$
4. $\omega_{\mathrm{R}} = 2449\,\mathrm{rad/s} \Rightarrow f_{\mathrm{R}} = 390\,\mathrm{Hz}$

Going further: Newton's law for each inertia with shaft torque $T_S = k(\theta_M - \theta_L) + c(\omega_M - \omega_L)$ gives the full transfer functions; a gear i reflects J_L/i^2, k/i^2.

Refresher: decibels (dB)

Refresher: decibels

Gains span many orders of magnitude, so we use a logarithmic scale:

$$G_{\text{dB}} = 20 \log_{10} |G|$$

Multiplying gains $\Rightarrow$ *adding* dB.

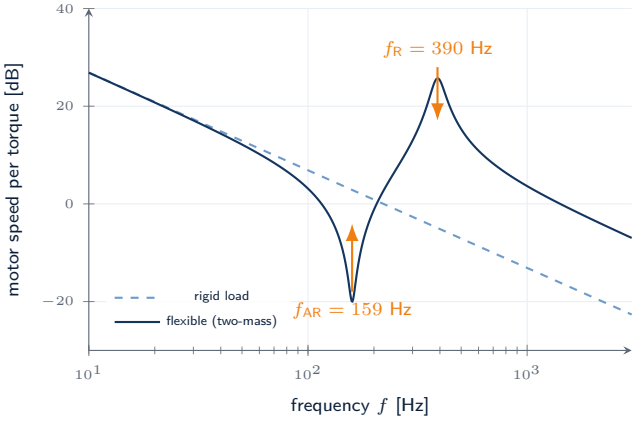
- 0 dB = gain 1 (signal passes unchanged).
- Positive dB = amplified; negative dB = attenuated.
- A **Bode plot**: gain in dB versus frequency on a log axis.

Gain $\lvert G \rvert$	dB
100	$+40$
10	$+20$
2	$\approx +6$
1	0
0.5	≈ -6
0.1	-20
0.01	-40

Example

A resonance peak of $+12$ dB $\approx$ gain $\times 4$ ($6 + 6$ dB $= 2 \times 2$).

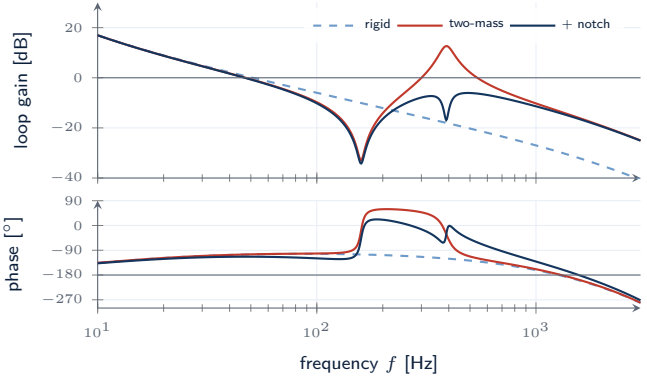
Frequency response of the two-mass system



- Low f: motor and load move as one body.
- Dip at f_{AR}, sharp **peak** at f_R.
- Above f_R: only the motor moves, the gain is $1 + R = 6$ times ($\approx$15.6 dB) higher.
- Large inertia ratio $R \Rightarrow$ bigger jump, harder tuning; keep $R \lesssim 5$–10.

Bicycle-gear analogy: a gearbox makes the load "feel" lighter to the motor (J_L/i^2), which reduces R.

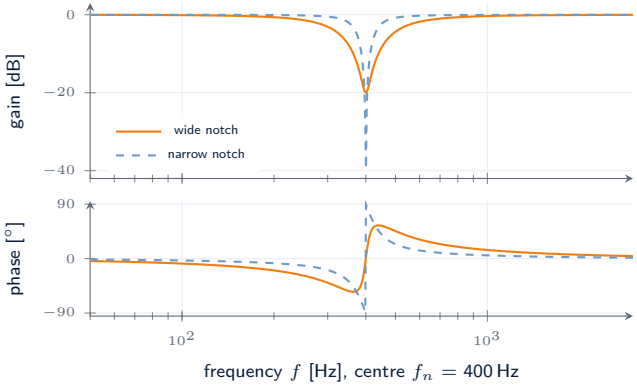
What the velocity loop sees



Velocity loop tuned for 50 Hz bandwidth, load from the example.

- Red: the resonance lifts the loop gain **above 0 dB** at 390 Hz ⇒ the axis rings.
- Blue: a **notch filter** at 390 Hz pushes the peak back down to −17 dB.
- Price: a little phase lost at 50 Hz (about 9°).

The notch filter: cutting out one frequency



- A **notch** (band-stop) filter removes a narrow band around f_n and passes everything else (0 dB).
- Place f_n on the resonance f_R.
- **Depth**: how much it cuts. **Width**: how many Hz it covers.
- Wide = robust if f_R drifts, but costs more phase at low f.

Going further: the biquad form $N(s) = \dfrac{s^2 + 2\zeta_z\omega_n s + \omega_n^2}{s^2 + 2\zeta_p\omega_n s + \omega_n^2}$, depth ζ_z/ζ_p, and its Tustin discretisation.

Setting up a notch: a practical recipe

1. Measure the frequency response at low gain (sine sweep).
2. Read f_R from the peak; check it at different loads and positions.
3. Set the notch centre $f_n = f_R$, depth 10 dB to 20 dB.
4. Make it wider if f_R moves by more than $\pm 10\,\%$.
5. Raise the velocity gain again and re-check the margins.

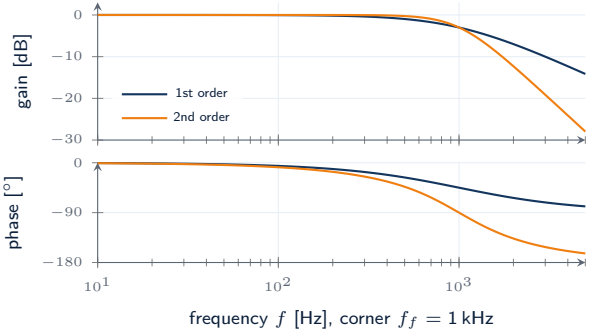
Rule of thumb

Keep the notch at least 3–5 times above the loop bandwidth: $f_n \gtrsim (3\text{–}5)\, f_c$. Example: $f_c = 50$ Hz, $f_n = 390$ Hz (ratio 7.8): only $\approx 9^\circ$ phase lost.

If the resonance is too close to f_c: stiffen the mechanics or lower f_c; no filter can fix it.

Going further: an **adaptive notch** estimates f_R online (e.g. FFT peak of the torque command) and moves f_n when the payload changes.

Low-pass torque filters



- A **low-pass filter** (LPF) passes slow signals and damps fast ones — like an RC circuit.
- Good against **broad** high-frequency noise: encoder steps, current ripple, many small modes.
- At 390 Hz a 1 kHz LPF cuts only 0.6 dB: it *cannot* fix a resonance near f_c. A notch can.
- Typical corner: $f_f \approx (5\text{–}10)\, f_c$.

Check your understanding: resonance

Check your understanding

1. The load inertia J_L is doubled. Does f_{AR} go up or down, and by what factor?
2. The loop bandwidth is 50 Hz and the resonance is at 390 Hz. Why use a notch rather than a low-pass filter?
3. A Bode plot shows a peak of $+20$ dB. By what factor is the gain amplified there?

Discuss in pairs for 2 minutes, then compare.

Summary and key terms: resonance and filters

- A flexible drive train is a mass–spring system with a sharp resonance.
- $f_{\mathrm{AR}} = \frac{1}{2\pi}\sqrt{k/J_L}$; f_{R} is $\sqrt{1+R}$ times higher.
- The resonance peak can push the loop gain above 0 dB $\Rightarrow$ ringing.
- **Notch** for one narrow peak; **low-pass** for broad high-frequency noise.
- Fix the mechanics first, filter second.

Key terms

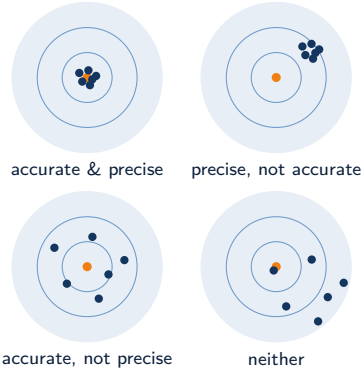
natural frequency rate of free oscillation
compliance springiness of a part
inertia ratio $R = J_L/J_M$
dB $20\log_{10}$ of a gain
notch narrow band-stop filter
low-pass passes slow, damps fast

[Ellis, 2012]; [Franklin, Powell & Emami-Naeini, 2019].

Session 2

Sensors and measurement principles (4.1)

Refresher: accuracy versus precision



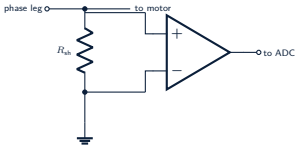
Refresher: the dartboard picture

Accuracy: how close the darts land to the bull's-eye (true value).
Precision: how tightly the darts cluster (repeatability).

Resolution	smallest detectable change (1 count)
Range	min. to max. measurable value
Linearity	deviation from a straight line

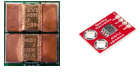
High resolution $\neq$ high accuracy: a ruler with 0.1 mm marks printed 2 mm off is fine-grained but wrong.

Current measurement: shunt, Hall and fluxgate



Example: shunt

1. $R_{sh} = 2\,\text{m}\Omega$, $i = 20\,\text{A}$
2. $u = R_{sh}\, i = 40\,\text{mV}$ (tiny: amplify $\times 20 \rightarrow 0.8\,\text{V}$)
3. Loss at 10 A: $P = R I^2 = 0.2\,\text{W}$

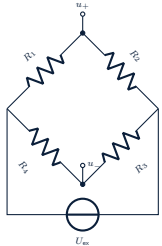


SMD shunts: wdwd, CC BY-SA 4.0; Hall IC: SparkFun, CC BY 2.0; via Wikimedia Commons

	Shunt	Hall	Fluxgate
Principle	$u = Ri$	field → voltage	zero-flux sensing
Isolated	no	yes	yes
Accuracy	high	medium	highest
Cost	lowest	medium	highest

- **Hall sensor**: the current's magnetic field bends charge carriers ⇒ small voltage $\propto i$.
- **Isolation**: no electrical connection between the high-voltage motor side and the logic.
- An offset error in the current gives torque ripple in the motor.

Force: strain gauges and load cells







Foil gauge: Pleriche, CC BY-SA 4.0; load cell: Daraceleste, CC BY-SA 4.0; via Wikimedia Commons

- A **strain gauge** is a thin foil resistor glued on metal: stretch it and R rises.
- Changes are tiny, so we use a **Wheatstone bridge** (left) to measure them.

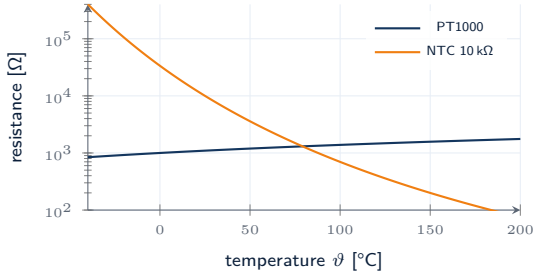
$$\frac{\Delta R}{R} = k_{\mathrm{GF}}\,\varepsilon \qquad (k_{\mathrm{GF}} \approx 2,\ \varepsilon = \text{strain} = \Delta L/L)$$

Example: full bridge

1. $\varepsilon = 10^{-3}$, $U_{\mathrm{ex}} = 5\,\mathrm{V}$, 4 active gauges
2. $u_{\mathrm{out}} \approx k_{\mathrm{GF}}\,\varepsilon\,U_{\mathrm{ex}} = 2 \cdot 10^{-3} \cdot 5\,\mathrm{V} = 10\,\mathrm{mV}$
3. Millivolts $\Rightarrow$ needs a low-noise amplifier

A **load cell** is a metal body with a bridge of gauges, rated in mV/V.

Temperature: PT100/PT1000, NTC, thermocouple, KTY







Left: wire-wound Pt100, Harke, CC BY-SA 3.0; right: type-K thermocouple, Harke, public domain; both via Wikimedia Commons

$$R \approx R_0\,(1 + \alpha\,\vartheta), \quad \alpha \approx 0.0039\,\mathrm{K}^{-1}$$

PT100: $R_0 = 100\,\Omega$ at 0 °C $\Rightarrow \approx 139\,\Omega$ at 100 °C.

- **PT100/1000** (platinum): nearly linear, accurate.
- **NTC**: resistance falls steeply with ϑ; sensitive, non-linear.
- **Thermocouple**: two metals give $\approx 41\,\mu\mathrm{V/K}$; for high ϑ.
- **KTY**: silicon sensor inside motor windings.

Check your understanding: sensors

Check your understanding

1. A scale shows 10.2, 10.2, 10.3 kg for a true 10.0 kg mass. Is it accurate, precise, both or neither?
2. A 1 mΩ shunt carries 30 A. What voltage does it give, and how much power does it dissipate?
3. Why do we use a bridge with four strain gauges instead of measuring one gauge directly?

Summary and key terms: sensors

- Accuracy (right on average) $\neq$ precision (repeatable) $\neq$ resolution (fine steps).
- Current: shunt (cheap, Ohm's law), Hall (isolated), fluxgate (most precise).
- Force: strain gauges in a bridge give millivolts.
- Temperature: PT for accuracy, NTC for sensitivity, thermocouple for heat.

Key terms

accuracy closeness to the true value
precision spread of repeated readings
resolution smallest visible change
shunt small precision resistor
strain relative stretch $\Delta L/L$
isolation no galvanic connection

[Fraden, 2016]; IEC 60751.

Session 3

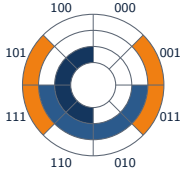
Encoder and resolver (4.2)

Refresher: binary numbers and Gray code

Refresher: binary

Each bit is a power of 2: $101_2 = 1{\cdot}4 + 0{\cdot}2 + 1{\cdot}1 = 5$.
N bits give 2^N values: 3 bits $\rightarrow$ 8, 12 bits $\rightarrow$ 4096, 23 bits $\rightarrow$ 8 388 608.

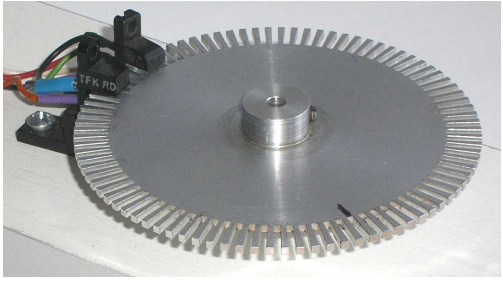
Value	Binary	Gray
2	010	011
3	**011**	010
4	**100**	110
5	101	111



3-bit Gray-code disk

- Binary $3 \rightarrow 4$: **all 3 bits** flip; if they flip at slightly different moments the reader may see a wrong value.
- **Gray code**: only **one bit** changes per step $\Rightarrow$ no false readings.

Optical incremental encoders



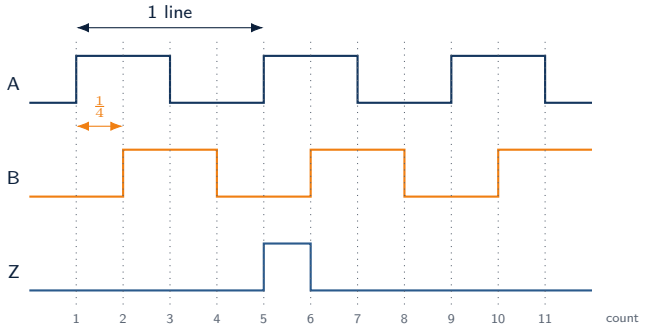
Rrudzik, CC BY-SA 4.0, via Wikimedia Commons

- An LED shines through a disk with N slits (lines) onto photodiodes.
- Outputs: square waves **A** and **B** (shifted by a quarter period) and an index pulse **Z** once per revolution.
- The drive counts pulses: position *relative* to switch-on $\Rightarrow$ **homing** needed.

Example: pulse frequency

1. $N = 2500$ lines, $n = 6000\,\text{rpm} = 100\,\text{rev/s}$
2. $f_A = N \cdot n = 2500 \cdot 100 = 250\,\text{kHz}$

Quadrature signals and $\times 4$ counting

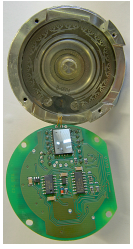


- Count **every edge** of A and B: 4 counts per line $\Rightarrow 4N$ counts/rev.
- **Direction**: if A changes before B we turn forward, otherwise backward.
- Both changing at once = a lost count $\Rightarrow$ error flag.

Example

$N = 2500$: $4N = 10\,000$ counts/rev $\Rightarrow 360^\circ / 10\,000 = 0.036^\circ$ per count.

Absolute encoders and serial protocols

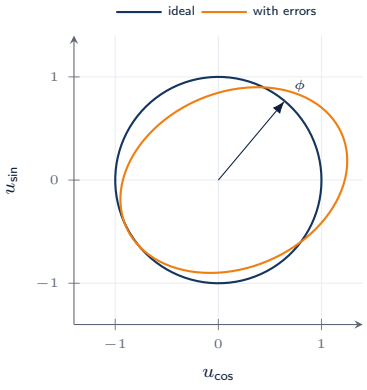


13-track Gray-code encoder, Mike1024, public domain, via Wikimedia Commons

- **Absolute**: every angle has its own code (many tracks, Gray-coded) $\Rightarrow$ position known at switch-on, **no homing**.
- **Single-turn**: e.g. 23 bit $= 8\,388\,608$ positions per rev. **Multi-turn**: also counts whole revolutions.
- Position is sent as a digital message over a serial line, protected by a checksum (CRC).

Protocol	Features
SSI	simple, one direction, no CRC by default
BiSS-C	open, two-way, CRC
EnDat 2.2	two-way, CRC, diagnostics
One-cable	data inside the motor power cable

Sin/cos encoders and interpolation



- Instead of square waves, the tracks give a **sine** and a **cosine** per line.
- Count whole periods m (coarse), then find the angle *inside* one period from the two voltages (fine):

$$\phi = \operatorname{atan2}(u_{\sin}, u_{\cos}), \qquad \theta = \frac{2\pi}{N}\left(m + \frac{\phi}{2\pi}\right)$$

- Example: $N = 2048$ periods, each split into 2^{12} steps $\Rightarrow 2^{23}$ counts/rev.
- Offset or amplitude errors distort the circle into an ellipse $\Rightarrow$ angle errors.

Resolver: a rotating transformer

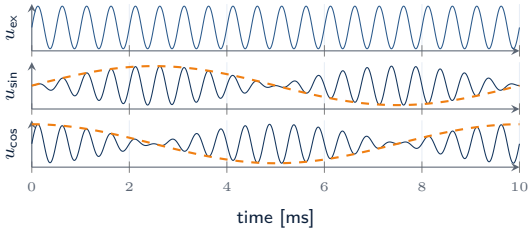
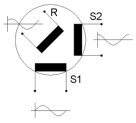


Illustration: carrier 2 kHz, rotor turning steadily.

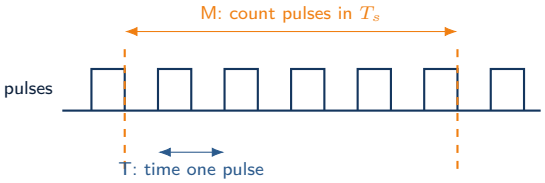


Hasee, public domain, via Wikimedia Commons

- Rotor coil fed with a sine **carrier** u_{ex}.
- Two stator coils at 90° pick up $\sin\theta$ and $\cos\theta$ times the carrier.
- The angle is in the **envelopes** (dashed).
- No electronics in the motor: robust to heat, shock, vibration.

Going further: a **resolver-to-digital converter** (RDC) demodulates the signals and uses a small tracking control loop to output angle and speed.

Speed from an encoder and the low-speed problem



- **M-method**: count pulses in a fixed time T_s — good at high speed.
- **T-method**: time between two pulses — good at low speed.
- Modern drives combine both (M/T, time-stamped edges).

Worked example: M-method

1. $4N = 10\,000$ counts/rev, $T_s = 125\,\mu s$
2. One count per sample $= \dfrac{1}{10\,000 \cdot 125 \cdot 10^{-6}} = 0.8$ rev/s
3. $= 48$ rpm: the smallest speed step
4. At 60 rpm the reading jumps between 48 and 96 rpm!

Remedies: more counts/rev (23-bit encoder: 0.057 rpm steps), longer windows, filtering (Session 4.4).

Check your understanding: encoders

Check your understanding

1. A 1000-line incremental encoder is decoded $\times 4$. How many counts per revolution, and how many degrees per count?
2. Why does an absolute encoder disk use Gray code instead of plain binary?
3. After a power failure, which encoder type needs a homing run: incremental or absolute? Why?

Summary and key terms: encoder and resolver

- Incremental A/B/Z: count $4N$ edges per rev, homing required.
- Absolute: Gray-coded disk, serial protocol with CRC, no homing.
- Sin/cos: fine angle inside a line from atan2.
- Resolver: robust analogue transformer sensor.
- Speed from counts is coarse at low speed.

Key terms

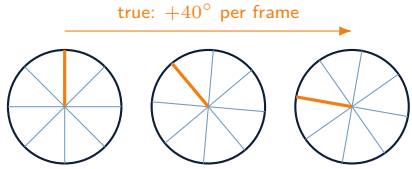
quadrature two signals 90° apart
homing finding a known zero position
Gray code one bit changes per step
interpolation angle inside one line
resolver rotary transformer sensor
CRC checksum to detect bit errors

[Krishnan, 2010]; ADI / TI application notes.

Session 4

Sampling and digitisation (4.3)

Refresher: sampling and the wagon-wheel effect



spokes repeat every 45°: the eye sees -5° per frame $\Rightarrow$ wheel seems to turn **backwards**

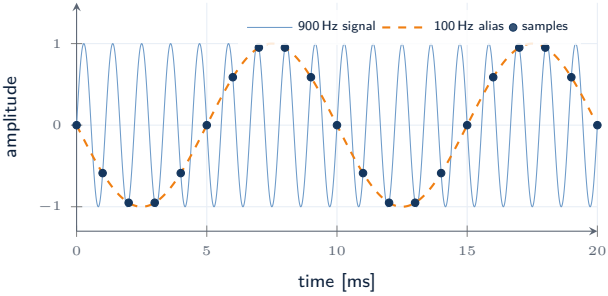
Refresher: sampling

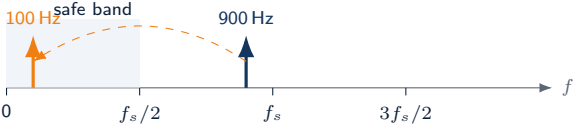
A computer only sees the signal at instants $t = kT_s$; the **sampling frequency** is $f_s = 1/T_s$. Like a film camera taking 24 pictures per second.

Nyquist: $f_{\text{signal}} < \dfrac{f_s}{2}$

- Faster signals are not lost, they are **disguised** as slower ones: **aliasing**.
- Same as a wagon wheel in a film or a strobe light.

Aliasing in a measured signal



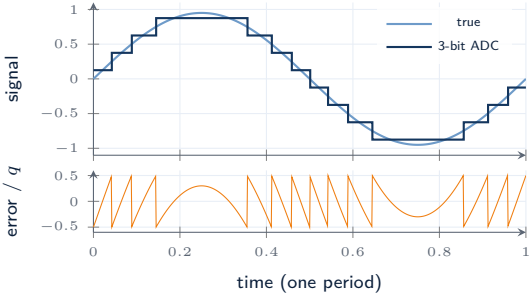


Example

1. $f_s = 1\,\text{kHz}$, so $f_s/2 = 500\,\text{Hz}$
2. Signal at 900 Hz $>$ 500 Hz: too fast
3. It appears at $|900 - 1000| = 100\,\text{Hz}$

- Rule: $f_{\text{alias}} = |f - k\,f_s|$, nearest whole k.
- After sampling, the fake 100 Hz looks exactly like a real one: **no digital filter can remove it**.

Quantisation: turning a voltage into a number



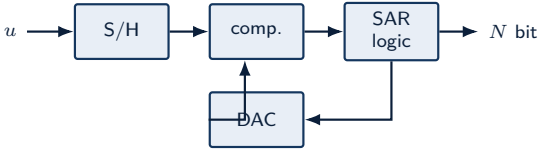
- An N-bit **ADC** (analogue-to-digital converter) has 2^N steps; step size $q = \text{range}/2^N$.
- Rounding gives an error of at most $\pm q/2$: **quantisation noise**.

$$\text{SNR} \approx 6.02\,N + 1.76 \text{ dB}$$

Example

1. Current range -50 to 50 A $= 100$ A, 12 bit
2. $q = 100/4096 = 24.4$ mA
3. SNR $\approx 6 \cdot 12 + 1.8 = 74$ dB; each extra bit $+6$ dB

ADC types for drives: SAR and sigma-delta



SAR

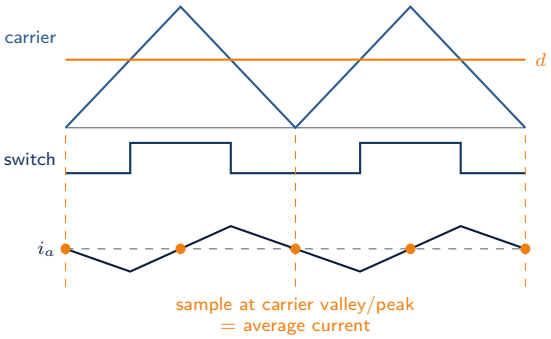
Is it above half range? Yes → bit 1. Now above $3/4$? ... One bit per guess: 12 guesses for 12 bits.

	SAR	Sigma-delta
Resolution	12–16 bit	16–24 bit
Delay	very short	longer
Typical use	phase currents	load cells, isolated current

- **S/H** (sample-and-hold) freezes the voltage during conversion.
- **Sigma-delta**: a very fast 1-bit converter, averaged to many bits.

Going further: $\Sigma\Delta$ modulators use oversampling and noise shaping; a sinc^3 decimation filter turns the bit stream into words.

Sampling in step with the PWM, and delay



- The PWM makes the current ripple (Day 4).
- Sampling exactly in the middle of a switching pattern gives the **average** current: no ripple, no aliasing.
- Every delay costs phase:

$$\Delta\varphi = 360^\circ \cdot f \cdot T_d$$

Example

1. $f_s = 16\,\text{kHz}$, delay $T_d = 1.5\,T_s = 94\,\mu\text{s}$
2. At $f = 1\,\text{kHz}$: $360{\cdot}1000{\cdot}94{\cdot}10^{-6} = 34^\circ$

Check your understanding: sampling

Check your understanding

1. A 9 kHz vibration is sampled at $f_s = 10$ kHz. At which frequency does it appear in the data?
2. How much better is the ideal SNR of a 16-bit ADC than a 12-bit one?
3. Why must the anti-aliasing filter be analogue, i.e. placed *before* the ADC?

Summary and key terms: sampling and digitisation

- Sample faster than twice the highest frequency (Nyquist).
- Too-fast content folds back as a fake slow signal (aliasing).
- N bits: step $q = \text{range}/2^N$, SNR $\approx 6N$ dB.
- SAR for speed, sigma-delta for resolution.
- Sample in sync with the PWM; every delay costs $360^\circ f T_d$.

Key terms

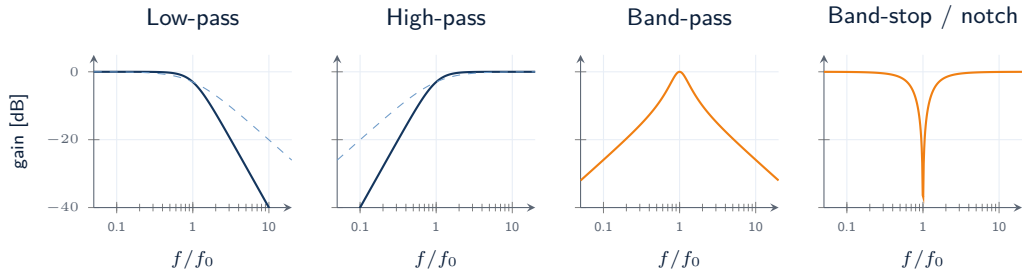
sampling frequency $f_s = 1/T_s$
Nyquist frequency $f_s/2$
aliasing fast signal disguised as slow
ADC analogue-to-digital converter
quantisation rounding to steps q
SNR signal-to-noise ratio in dB

[Oppenheim & Schafer, 2010]; [Franklin, Powell & Workman, 1998].

Session 5

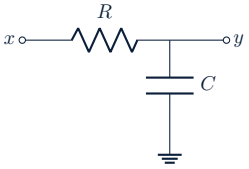
Signal processing and filters (4.4)

Filter types at a glance



Type	Passes	Typical use in a drive
Low-pass	slow signals	smoothing noisy speed, anti-aliasing
High-pass	fast signals	removing offset / slow drift
Band-pass	one band	picking out one vibration frequency
Band-stop	all but one band	suppressing a resonance (notch)

From an RC low-pass to one line of code



- You know the RC circuit: y follows x with time constant $\tau = RC$; corner $f_f = 1/(2\pi\tau)$.
- In software: each sample, move y a fraction α of the way towards x:

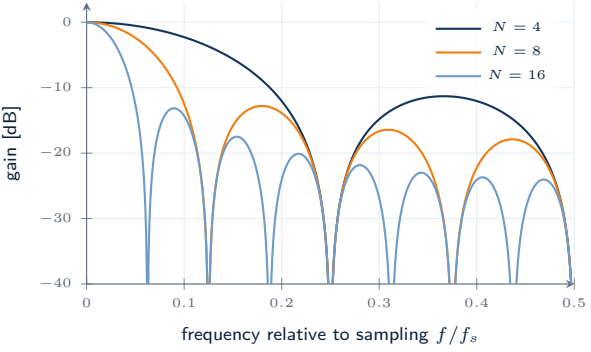
$$y[k] = y[k-1] + \alpha\,\big(x[k] - y[k-1]\big), \qquad \alpha \approx \frac{T_s}{\tau}$$

Worked example

1. Want $f_f = 500\,\text{Hz} \Rightarrow \tau = 1/(2\pi \cdot 500) = 318\,\mu\text{s}$
2. $f_s = 16\,\text{kHz} \Rightarrow T_s = 62.5\,\mu\text{s}$
3. $\alpha \approx 62.5/318 = 0.196$ (exact: $1 - e^{-T_s/\tau} = 0.178$)
4. Code: `y += 0.178f*(x - y);`

Every low-pass filter is also a **delay** of about τ (here 0.3 ms, 5 samples). Inside a loop that costs phase margin.

Moving average: the simplest digital filter



$$y[k] = \frac{1}{N} \sum_{i=0}^{N-1} x[k-i]$$

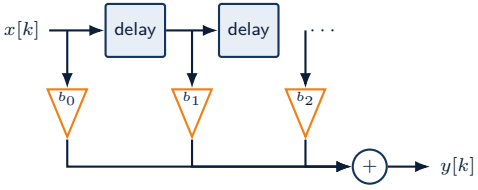
- Average of the last N samples.
- Cancels a disturbance **exactly** if its period fits the window: zeros at $f = f_s/N, 2f_s/N, \ldots$
- Delay: half the window, $\frac{N-1}{2} T_s$.
- Weak damping between the zeros.

Example

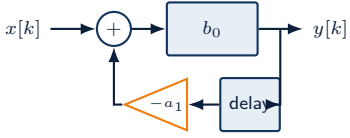
$f_s = 16\,\text{kHz}$, $N = 16$: cancels 1 kHz, 2 kHz, ...

FIR versus IIR filters

FIR: weighted past *inputs*



IIR: also feeds back past *outputs*

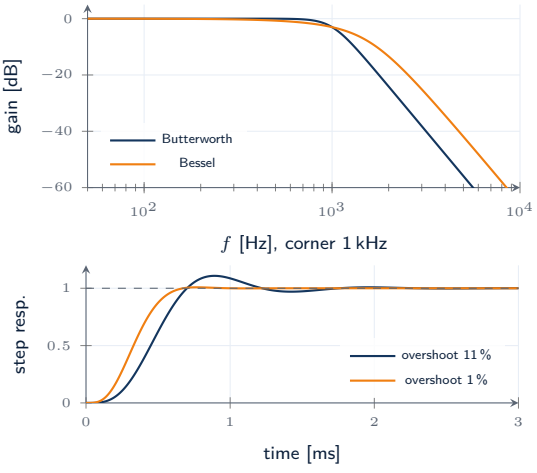


	FIR	IIR
Memory	past inputs	inputs + outputs
Stable	always	if designed so
Effort	many terms	few terms
Delay	larger	small

- Moving average = FIR. RC-type filter = IIR.
- Inside control loops **IIR** usually wins (low delay).

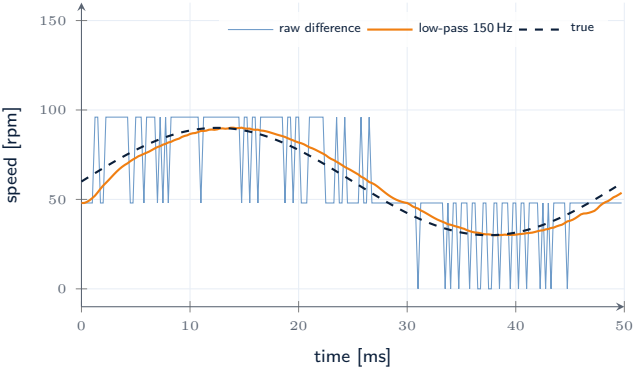
Going further: higher-order IIR filters are built as cascades of second-order sections ("biquads", Direct Form II transposed) to avoid rounding problems.

Butterworth vs. Bessel: sharp or clean?



- Both are 4th-order low-pass filters with the same corner.
- **Butterworth**: flattest pass band, steeper cut-off (-38 dB at $3\times$ corner) — but **rings** on a step.
- **Bessel**: gentler cut-off (-25 dB), but keeps the **shape** of the signal: clean step.
- Use Butterworth to remove noise, Bessel when the waveform matters (transients).

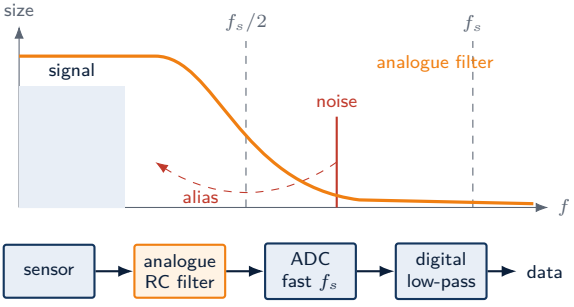
Speed from position: differentiating amplifies noise



Simulation: 10 000 counts/rev, $T_s = 125\,\mu s$, true speed 60 ± 30 rpm.

- Speed $\approx \dfrac{\theta[k] - \theta[k-1]}{T_s}$: a difference of two rounded numbers.
- Dividing a one-count step by a tiny T_s gives big jumps (0, 48, 96 rpm).
- Differentiation boosts fast changes: $+20$ dB per decade of frequency.
- The low-pass smooths it but adds **delay** ($\approx$1.5 ms).
- Better: more resolution or an **observer** (Day 3).

Anti-aliasing: analogue before the ADC, digital after



- Only an **analogue** filter in front of the ADC can stop aliasing (see wagon wheel).
- Sampling much faster than needed (**oversampling**) lets a simple RC do the job; the sharp filtering is then done digitally.
- The analogue filter also adds delay: keep its corner ≥ 5–$10\times$ the loop bandwidth.

Check your understanding: filters

Check your understanding

1. A moving average with $N = 8$ runs at $f_s = 8\,\text{kHz}$. Which frequencies does it cancel exactly?
2. Why is a speed signal computed by differencing encoder counts so noisy at low speed?
3. You need a filter inside a fast velocity loop. FIR or IIR — and why?

Summary and key terms: signal processing and filters

- Every filter changes the gain *and* adds delay; in a loop both matter.
- First-order low-pass = digital RC: one line of code.
- Moving average (FIR) cancels known periodic disturbances.
- IIR: few terms, low delay; FIR: always stable, more delay.
- Differentiation amplifies noise; aliasing is stopped only before the ADC.

Which filter?

narrow resonance	notch
broad noise	low-pass
offset / drift	high-pass
known periodic dist.	moving avg.
anti-aliasing	analogue LPF

FIR finite impulse response; **IIR** infinite impulse response; **difference equation** filter as a sample-by-sample formula.

Session 6

Communication and system integration (4.5)

Refresher: what is a network packet?



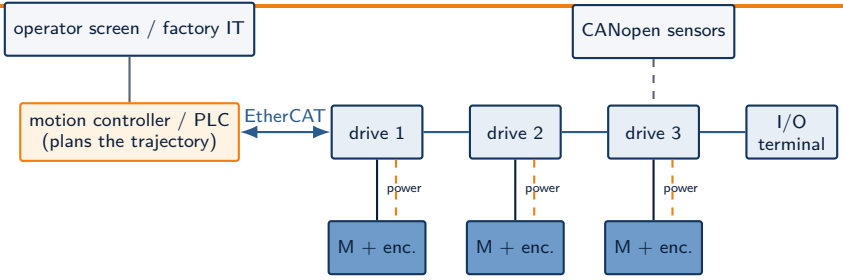


Refresher: packets

A **packet** (or **frame**) is a short block of bits sent over a cable, like a letter: an envelope with addresses, the letter inside, and a seal that shows tampering.

- **Bit rate**: bits per second on the wire.
- **Cycle time**: how often fresh packets are exchanged.
- **Latency**: how old the data is when it arrives.
- **Fieldbus**: a network built for machines, with guaranteed timing.

Communication architecture of a multi-axis machine



Link	Data	Cycle	Example
Inside the drive	currents, angle	31 µs to 125 µs	SPI, BiSS, EnDat
Drive ↔ controller	set-points, actual values	0.125 ms to 4 ms	EtherCAT, PROFINET, CANopen
Controller ↔ factory	recipes, diagnostics	10 ms to 1000 ms	Ethernet TCP/IP, OPC UA

CAN and CANopen

- **CAN**: two twisted wires shared by all nodes, $120\,\Omega$ resistor at each end.
- Each message has an **identifier**; if two nodes send at once, the lower identifier wins (**arbitration**) — like a polite queue by priority.
- Up to 8 data bytes per frame, 1 Mbit/s over short cables.
- **CANopen** adds a common dictionary of parameters and fast process-data messages (**PDO**).

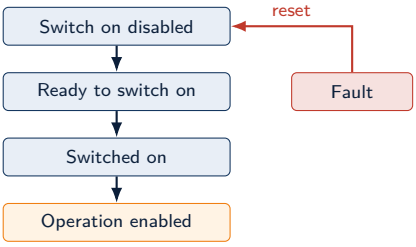


Worked example: how fast?

1. One full frame ≈ 120 bits $\Rightarrow \approx 120\,\mu s$ at 1 Mbit/s
2. 4 axes $\times$ 2 frames $+$ 1 sync frame $= 9$ frames
3. $9 \times 120\,\mu s \approx 1.1$ ms per cycle

$\Rightarrow$ fine for $\geq$1 ms cycles and few axes only.

CiA 402: a common language for drives



- Like a car: unlocked → ignition → engine on → driving. A fault drops it back.

- **CiA 402** is a **drive profile**: every vendor's drive understands the same commands and parameters.
- The controller writes a **control word**, the drive answers with a **status word**.
- Most used modes: **cyclic synchronous position / velocity / torque** (CSP, CSV, CST): the controller sends a new set-point every bus cycle.
- Same profile over CAN and EtherCAT.

Going further: the full state machine (8 states incl. quick stop and fault reaction), object indices 0x6040/0x6041, 0x607A, . . . , IEC 61800-7.

EtherCAT: processing on the fly



- Like a train passing stations: each drive **unloads** its set-points and **loads** its actual values while the frame rushes through (well under 1 µs per drive).
- A **working counter** in the frame tells the master that every drive answered.
- **Distributed clocks**: all drives share one time base and act at the same instant.

Example: 32 axes

1. $32 \times 12\,\text{B} = 384\,\text{B} + 52\,\text{B}$ overhead
2. ≈ 3500 bit at 100 Mbit/s ≈ 35 µs
3. + delays ≈ 30 µs $\Rightarrow$ fits a 100 µs cycle



EtherCAT slave controller IC: Allanchou1357, CC BY-SA 4.0, via Wikimedia Commons

Going further: distributed-clock details: propagation-delay measurement, drift compensation, SYNC0 events, sub-µs jitter.

Comparing interfaces and fieldbuses

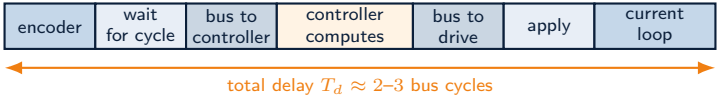
Interface	Speed	Typical cycle	Use
SPI	10 MHz to 50 MHz	every sample	chips on one board (ADCs)
RS-485	up to 10 Mbit/s	—	encoder links, Modbus
CAN / CANopen	1 Mbit/s	$\geq$1 ms	robust, cheap, slow
EtherCAT	100 Mbit/s	62.5 µs–4 ms	fast multi-axis motion
PROFINET IRT	100 Mbit/s	31.25 µs–ms	fast multi-axis motion

How to choose

Choose the bus by **cycle time**, **synchronisation**, number of axes and drive profile — not by raw bit rate alone.

Standards: ISO 11898 (CAN), IEC 61158 / 61784 (fieldbuses), IEC 61800-7 (drive profiles), TIA/EIA-485.

Latency budget: which loops may run over the bus?



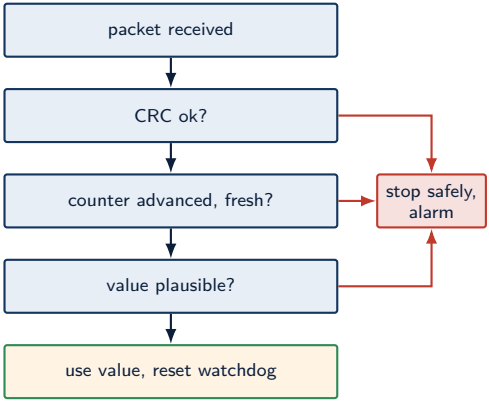
Worked example

1. Bus cycle 250 µs, $T_d = 2.5 \times 250 = 625$ µs
2. Position loop, $f_c = 30$ Hz: $360 \cdot 30 \cdot 625 \cdot 10^{-6} = 7^\circ$ ok
3. Velocity loop, $f_c = 200$ Hz: 45° too much

- Close the fast loops (current, velocity) **inside the drive**.
- Send only position set-points over the bus (CSP mode).
- Shower analogy: a long pipe (delay) makes temperature control slow — keep fast control close to the tap.

Is the data valid? CRC, counters, watchdogs, timestamps

- **CRC**: detects bits damaged on the way. But a correct value can still be *old*.
- **Counters**: a number that increases every cycle; a frozen counter = lost data.
- **Watchdog**: a timer that trips if no fresh data arrives in time.
- **Timestamp**: age of a value $= t_{\text{now}} - t_{\text{sample}}$.
- **Plausibility**: is the value possible? e.g. $i_a + i_b + i_c = 0$.



Check your understanding: communication

Check your understanding

1. What are the three parts of a network packet, and what is each for?
2. A bus has a total delay of 1 ms. How much phase does this cost a loop with $f_c = 100$ Hz? Would you close that loop over the bus?
3. A value arrives with a correct CRC but is five cycles old. Which mechanism catches this?

Summary and key terms: communication and integration

- Packets = header + data + CRC; buses exchange them every cycle.
- CAN/CANopen: robust, priority arbitration, $\geq$1 ms cycles.
- CiA 402: one command language for all drives.
- EtherCAT: one frame through all drives, 100 µs cycles, common clock.
- Delay costs phase: close fast loops inside the drive; check freshness of data.

Key terms

packet header + data + check
fieldbus machine network with fixed timing
arbitration who may send first
drive profile standard drive commands
latency age of data on arrival
watchdog timer that trips on missing data

ETG.1000 / IEC 61158; CiA 301, CiA 402; ISO 11898.

Session 7

Course wrap-up

Course wrap-up: five days, one servo axis

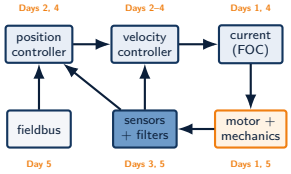
Day 1 Electrical machines & actuators; modelling

Day 2 System dynamics, identification, PID, stability

Day 3 Filters, discrete and advanced control; servo intro

Day 4 Drives, FOC, cascaded loops, tuning

Day 5 Resonance, sensors, sampling, filters, fieldbus



Today in one sentence

How good a servo can be is set by the mechanics (resonances) and the measurement chain (resolution, noise, delay) — the controller can only use what they allow.

Course wrap-up: ten principles to take home

1. Know your plant: motor *and* mechanics (Days 1, 5).
2. Model first, then measure and check (Day 2).
3. Feedback needs margins: keep a safe phase margin (Day 2).
4. Every filter and every sample adds delay (Days 3, 5).
5. Cascade loops: inner loops 3–10$\times$ faster (Day 4).
6. Field orientation makes the motor a torque source (Day 4).
7. Feed-forward does the work, feedback fixes errors (Day 4).
8. Fix resonances mechanically first, then notch / LPF (Day 5).
9. Sensor resolution, noise and delay limit stiffness (Day 5).
10. Synchronise, time-stamp and check all data (Day 5).

Thank you

Questions and discussion — and next time you meet a real machine, try to spot its resonance, its sensors and its bus.

Figure credits

All plots and diagrams not listed here are original drawings computed for this course (TikZ / pgfplots; data generated by `day5/data/gen_data.py`).

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- ISO 11898-1/-2 (CAN); CiA 301 and CiA 402 (CANopen, drive profile); IEC 61800-7 (drive profiles).
- IEC 61158 / IEC 61784 and ETG.1000 (EtherCAT); IEC 60751 (industrial platinum resistance thermometers).
- Application notes on resolver-to-digital conversion, sigma-delta current sensing and encoder interfaces: Analog Devices, Texas Instruments, Infineon.